Variability and acceleration in the growth of children's early vocabulary

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Children's early word learning is alternatively portrayed as either a process of gradual accumulation or as an inferential process that results in rapid ma be rectified? Here we present a statistical model that describes both the acceleration of children's vocabulary growth and its remarkable variability ac how changes in children's learning efficiency mean that younger children accumulate knowledge slowly while older children make rapid inferences. Fu estimate how much variation between children is due to differences in home language environment, a key policy-relevant question. Estimates from four long recent meta-analysis in suggesting XYZ% of total variance. Finally, we show that, in contrast to children, large language models are not well-described by accumulator dynamics and very limited variability across a number of dimensions. These results suggest ...

Early word learning; Bayesian data analysis; Large language models

Children's early language learning is often portrayed as a process of accumulation and gradual learning, with statistical learning, mutual exclusivity and input viewpoints highlighting the accumulation nature. In contrast, studies of processing and word mapping highlight the increases in efficiency that happen over the same period. How can these two views be rectified? One clue comes from examining the two key signatures of early vocabulary growth: variability and acceleration. What is the shape of this acceleration? And what is the source of variability, from children's input or otherwise?? We present a quantitative model that unify these views while explaining both the variability and acceleration of children's early language growth. The resulting model provides a good quantitative explanation for longitude vocabulary growth across large population of children. It also shows that by their increasing efficiency, and their ability of children, learning fundamentally different ways than large language, models, which show classic accumulator dynamics.

Children's early vocabulary grows very rapidly during early childhood, with large variation between children. The nature of this growth has theoretical, practical, and methodological importance.

From a theoretical perspective, growth mechanisms shed light on children's fundamental learning mechanisms and how they change across childhood. From a practical perspective, variation in growth is an important precursor — and predictor — of academic achievement. From a methodological perspective, vocabulary also affords a unique psychometric opportunity to study human variation because, unlike nearly all important human psychological metrics, there is a true zero point - a time before which children know no words. Thus, absolute levels of variability can be calculated.

One dominant viewpoint is that this growth can be characterized as a process of accumulation (McMurray 2007; Mitchell & McMurray, 2009; Hidaka, 2013; Mollica & Piantadosi, 2017; Kachergis et al., 2021). This pure accumulator view connects with a viewpoint in which the quantity of language that children hear has a strong causal role in the speed of their learning (Fernald papers; Bergeleson 1001). It also connects with the idea that large language models – whose learning scales in predictable, accumulator-like ways (Kaplan, Chinchilla, scaling laws) – might be good models for children's learning.

But important parts of this proposal have not been fleshed out. For example, how much variation is due to input quantity vs. other sources? Meta-analysis of this relationship suggests it's there but not that big (Coffey & Snedeker 2026). Could be quality as well, of course.

Furthermore, how does the accumulation process relate to the rapid growth of vocabulary? On some models, accumulation by itself can produce an exponential "vocabulary explosion" while the child's own learning abilities remain constant. On other accounts, however, the child's own abilities change over development. These changes could be due to the accumulation of language allowing bootstrapping (e.g., via mutual excusivity or the shape bias), they could be through faster processing (Fernald rich get richer), or they could be via general maturation (e.g., better memory, faster processing). Or all of these.

We address these questions by creating Bayesian data analysis models that connect proposals about the nature of the accumulation process with large-scale data about the longitudinal growth of children's vocabulary over time. Our key

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innovations are 1) that we model population variability between children, not just the central tendency and 2) that we directly connect quantitative estimates of language exposure to the rate of vocabulary growth.

Critically, we make use of new data resources, first connecting between vocabulary growth and data-driven estimates of the general range of language input quantity available to children (study 1), then to estimates of specific children's language input from BabyView and SEEDLings datasets (study 2), and finally to estimates of children's processing speed from Peekbank (study 3).

Overall, our analysis supports a view in which children's vocabulary growth is not a pure process of accumulation but rather one that accelerates rapidly over time. Further, although variation in input rate accounts for significant variation in learning speed, it is limited to approximately 10% of total variation across datasets.

Finally, we explore an application of the same framework to large language models. Making use of GPT-2 models trained on human developmental data, we show that – in contrast to children – these models do appear to function as pure accumulators. This analysis exposes a critical disanalogy between LLMs and human learners and suggests a potential route for understanding the vast "data gap" between children and large language models (Frank, 2023).

Here we ask whether these two signatures of vocabulary growth - variation and acceleration - are consistent with pure accumulator models, which are prevalent in the word learning literature and which characterize the scaling dynamics of large language models.

Model

We start with a simple accumulator model that merges ideas from item response theory and learning theory (McMurray, Kachergis). We begin with the Rasch model.

Child i produces word j has probability $(\eta_i j)$

$$P(w_{i,j} = 1 \mid \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}$$

This is an IRT model – specifically a Rasch model – where θ_i is child i's ability and δ_j is word j's difficulty. Following (**kachergis2021?**), we consider a simple accumulator model ("pure accumulator"), specifying that θ_i is not a single parameter for a child but instead a function of accumulation over time. In the simplest form:

$$\theta_i = log(r_i) + log(t)$$

where r_i is the child's rate of language innput (in tokens per month) and t is the child's age in months. In the exponential, this formulation produces a simple rate X time computation. In other words, ability increases over time. This form instantiates the basic hypothesis that the only thing that changes over time is more input, and the only source of variation between children is variation in rate.

This model does not have any way to capture variation across children. Thus, we augment this model by adding this model in two ways. First, we add variation related to accumulation rate:

$$\theta_i = log(\alpha_i) + log(r_i) + log(t)$$

In this formulation,

Second, we add a temporal acceleration component (an exponent):

$$\theta_i = log(\alpha_i) + log(r_i) + \kappa_i * log(t)$$

When exponentiated, this turns into

$$\alpha_i * r_i * t \exp \kappa_i$$

.

This final model instantiates the idea that children vary in their base level of variation (α) and their acceleration rate (κ).

As we will show, such a model fits a range of data vastly better than models without these terms.

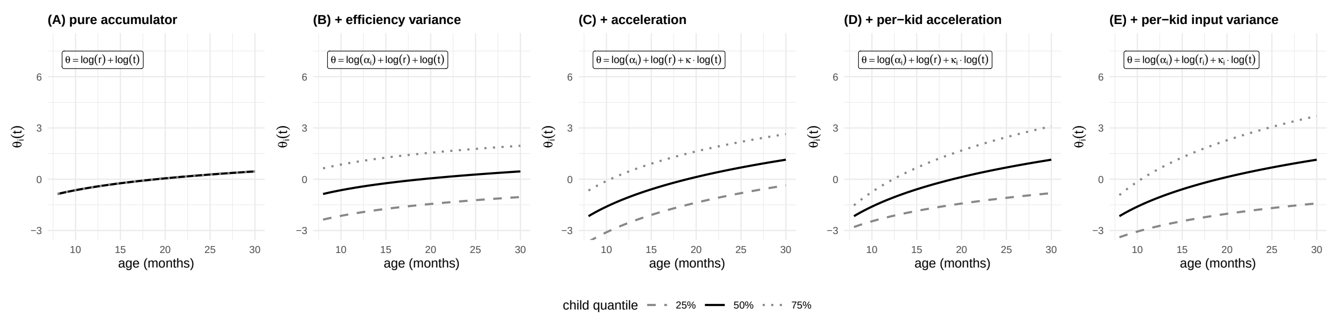


Figure 1

Model comparison. We used the standard reduction of the Rasch model to a mixed effects regression model (**jss_paper?**) to fit the set of models described above.

We fit these models to data on children's language production. Our primary data source is the MacArthur-Bates Communicative Development Inventories [MB-CDI; (**fenson1994?**);(**marchman2021?**)], a popular parent report instrument in which parents are asked to indicate which words their child produces and understand. Here we focus exclusively on production, which is used with older children and which tends to be a more reliable measure [(**frank2021?**);others]. We make use of data from Wordbank (**frank2017?**), an open repository of data from the MB-CDI. Since our efficiency variation model is not identifiable in cross-sectional data, we focus on those datasets that include substantial amounts of longitudinal data, which come from American English (N=XYZ), Norwegian (N=XYZ), Quebec French (N=XYZ), and Japanese (N=XYZ).

The efficiency change was the best fitting model across all four datasets (as well as two smaller ones; see SI), with AIC differences of XYZ–XYZ. Figure 1 shows the four primary model variants: pure accumulator models and models that include population-level efficiency change predict no variation (A, B). Allowing both efficiency (intercept) and efficiency change (slope) to vary across individuals results in substantially better fit.

Power-law growth models (in which growth in ability is a function of the logarithm of age) showed statistically better fits than exponential growth models (in which growth in ability is a linear function of age). Though these models were better from a statistical perspective (δAIC XYZ − XYZ), differences were small within the age range we examined (see Supplemental Table XYZ and Figure XYZ for all model fits).

Table 1

Panel	Dataset	Citation	N children	Admins/kid (median, range)	Age range	Input/recordings
Wordbank longitudinal	English (American)	(**CITE?**)	NA	NA	NA	—
Wordbank longitudinal	Norwegian	(**CITE?**)	NA	NA	NA	—
Wordbank longitudinal	French (Quebecois)	(**CITE?**)	NA	NA	NA	—
Wordbank longitudinal	Japanese	(**CITE?**)	NA	NA	NA	—
Input-observed (IO)	BabyView	(**CITE?**)	22	5 (1–8)	6–32 mo	Head-cam, n=5739 videos
Input-observed (IO)	SEEDLingS	(**CITE?**)	44	12 (6–13)	6–17 mo	LENA, n=525 day recordings
Input-observed (IO)	AM2018	(**CITE?**)	66	3 (1–3)	16–24 mo	LENA, n=126 recordings

Table 1

Panel	Dataset	Citation	N children	Admins/kid (median, range)	Age range	Input/recordings
Input-observed (IO)	FMW2013	(**CITE?**)	51	3 (1–3)	18–30 mo	LENA, n=51 recordings

The best-fitting modified accumulator model included large developmental changes in efficiency across all languages ($\beta = XYZ - XYZ$). This finding suggests that we can robustly reject pure accumulation models of early word learning. Instead, children get vastly more efficient over development.

This change in efficiency could come from a variety of sources, including general maturation of memory and attention, specific changes in word recognition or processing (**fernald1998?**; **frank2026?**), the increasing sophistication of inferential strategies for word learning [(**smith2002?**);(**markman1988?**);etc], or other sources.

As an illustration of this change in efficiency, Figure 2 shows a model-derived estimate of the average number of times a child would need to hear a word before producing it (See SI: Changes in Word Efficiency), using frequency estimates from the CHILDES Corpus (**mcwhinney2020?**; **sanchez2019?**). Across different word classes (e.g., nouns, verbs, closed class words), efficiency increases exponentially, such that early-learned words such as "ball" may be heard tens of thousands of times before they are produced, while later learned words such as "vitamin" might only be heard tens of times before being produced.

Estimating input. Under a range of plausible assumptions about variation in input, all children show supra-linear scaling of growth, even accounting for accumulation dynamics.

As discussed above, one important question is how much of variance is due to variance in rate. We explore this by asking about the parameter π,

$$\pi = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_r^2}$$

If $\pi > .5$, then the majority of variation between individuals is not due to pure accumulation rate.

In addition, 80-90% of between-child variation remains unexplained by input quantity. These conclusions are supported by analysis of longitudinal input data from BabyView (N=20) and SEEDLings (N=44) datasets, which contain child-specific estimates of input, and Peekbank (N=XYZ), which contains child-specific estimates of processing speed.

Thus, modeling human learning requires positing other sources of variability and acceleration. Sources of variability could include from variation in interactional input quality, genetic variation between children, environmental richness not captured by input, etc. Sources of acceleration could include increases in processing efficiency, learning-to-learn including mutual exclusivity, shape bias, syntactic bootstrapping, as well as domain general changes in perception, processing speed, memory, attention or other aspects of cognition. The key to understanding children's data efficiency may thus lie in better understanding when and how children's learning accelerates with maturation and experience.

We end by providing a quantitative comparison between within-child human learning and LLM scaling laws, showing that children's early language learning is strikingly different from LLM scaling: it shows both steeper scaling and greater variance in growth rate.

Discussion

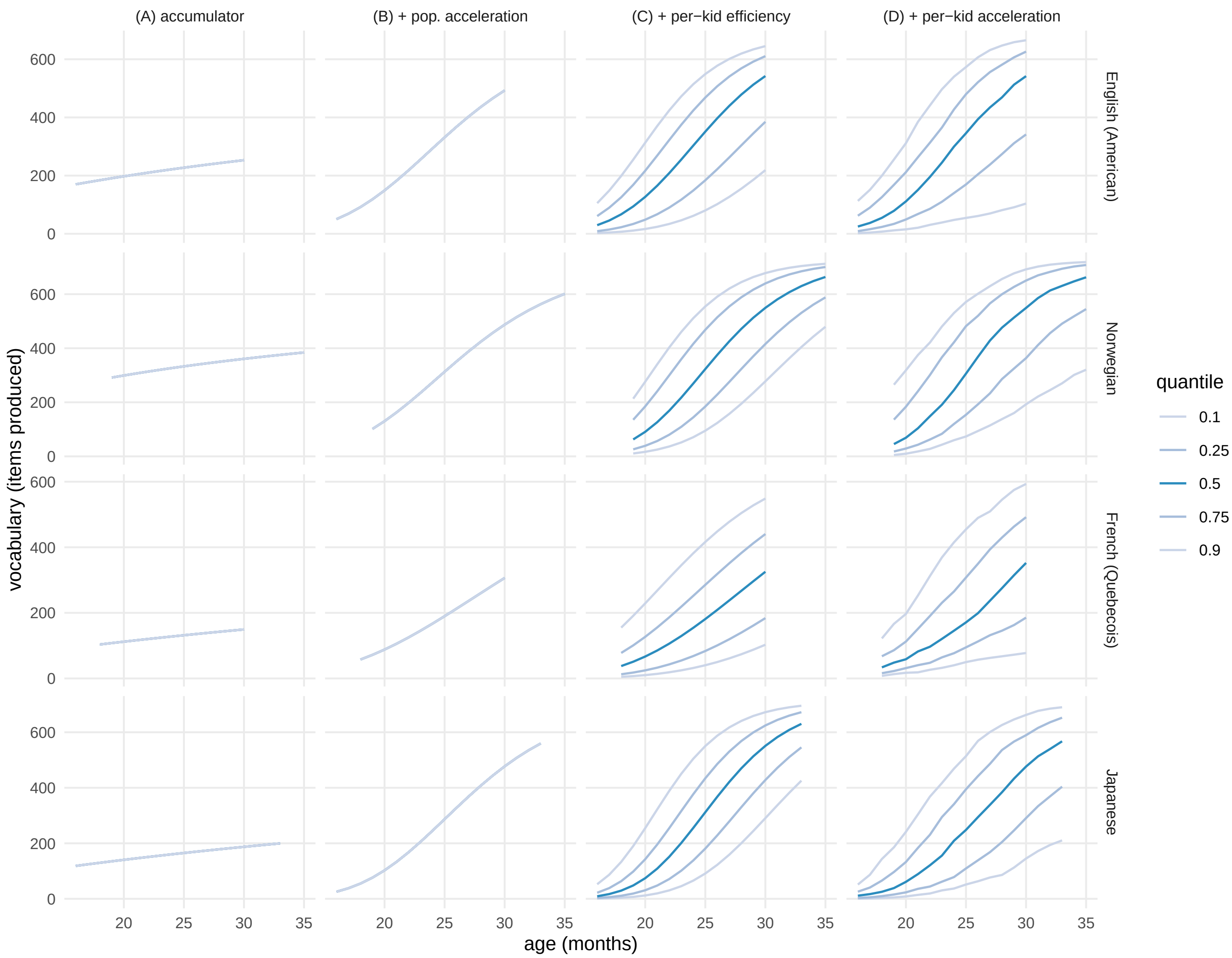
Coffey 2026 variance decomposition

Blingualism

International adoption

https://bergelsonlab.fas.harvard.edu/sites/g/files/omnuum3941/files/2025-01/meylan_bergelson_2021_ARL.pdf

Claude Code was used for literature review, code generation, visualization, and simulation infrastructure. All writing is original to the author.



(A) accumulator
(B) + pop. acceleration
(C) + per−kid efficiency
(D) + per−kid acceleration
English (American)
Norwegian
French (Quebecois)
Japanese
vocabulary (items produced)
age (months)
quantile
0.1
0.25
0.5
0.75
0.9

Figure 2

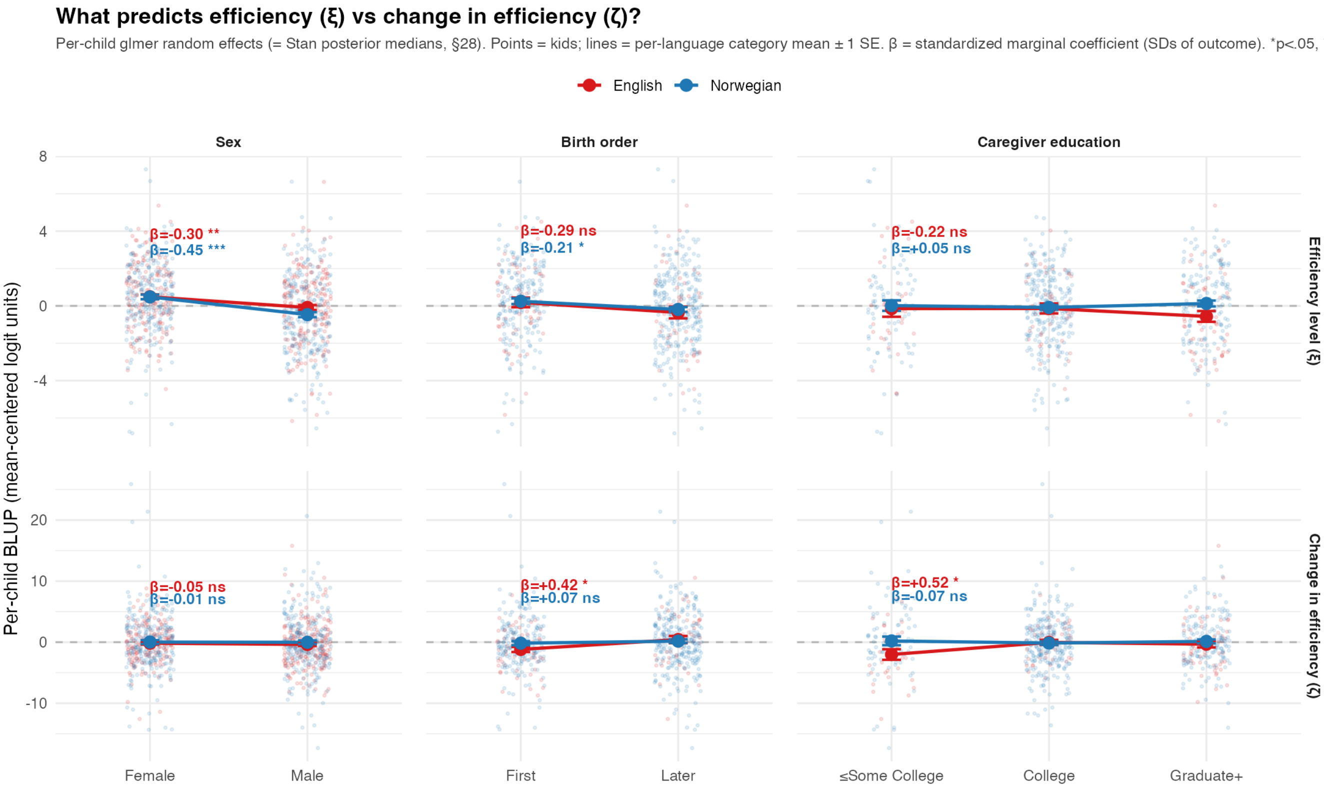


Figure 3

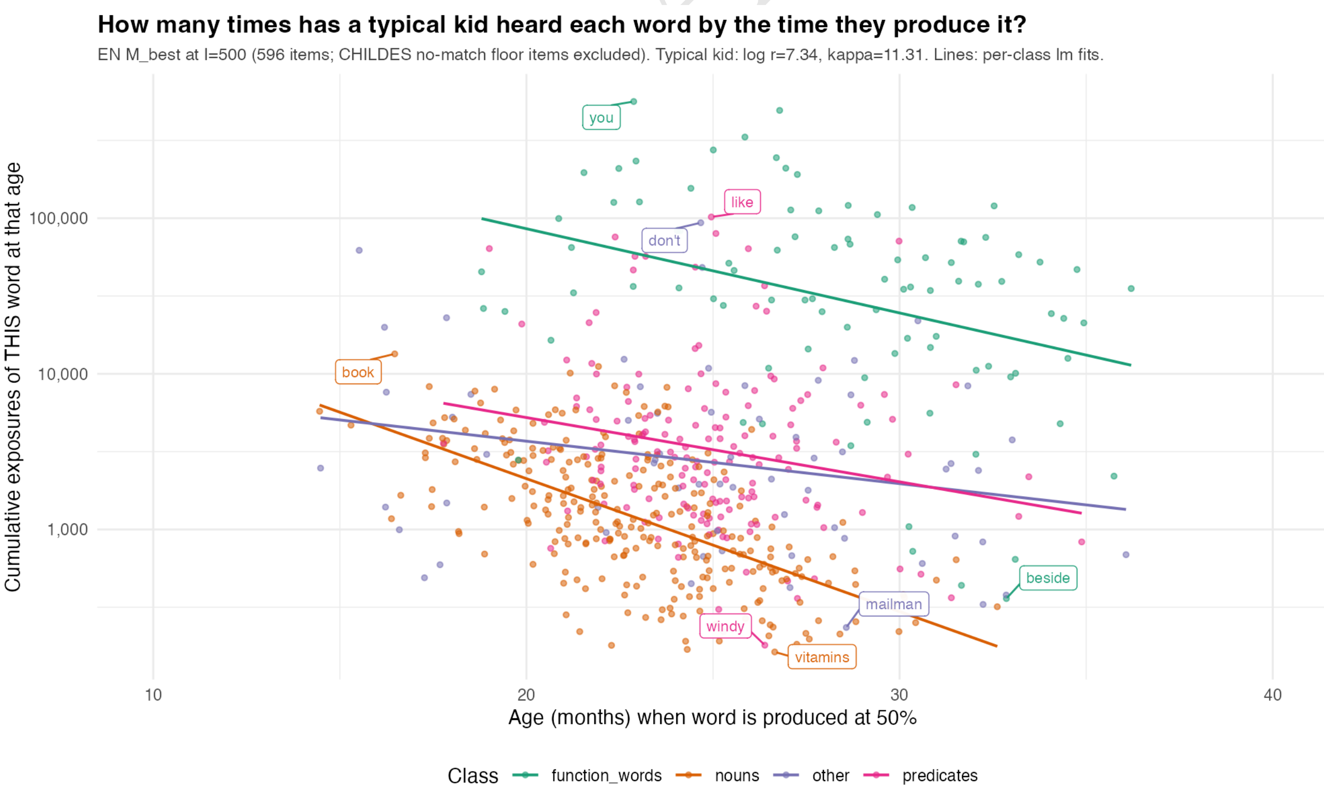


Figure 4

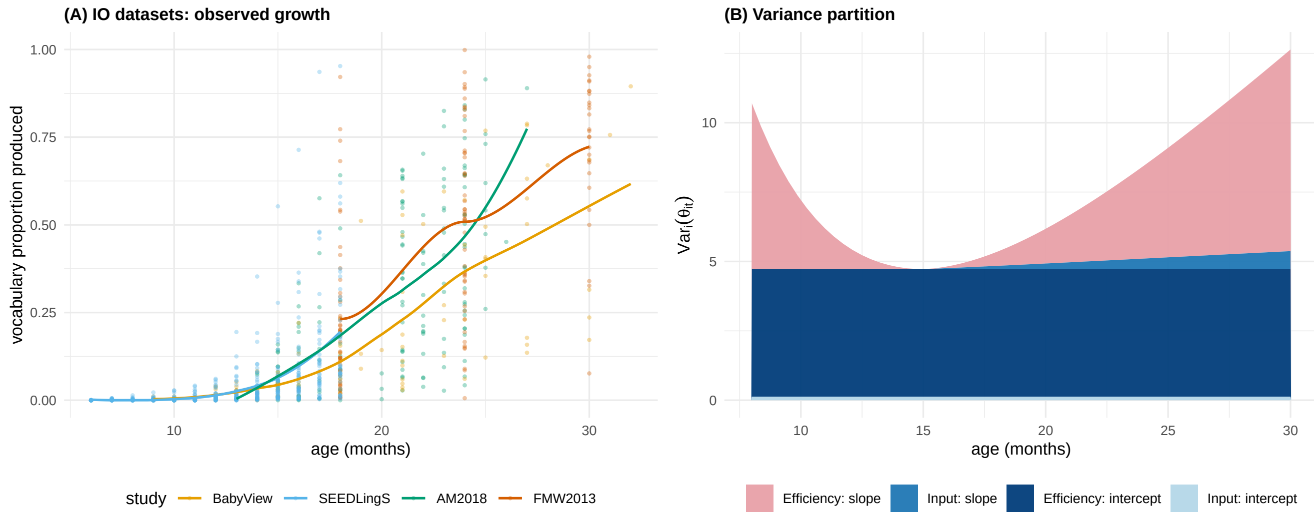


Figure 5

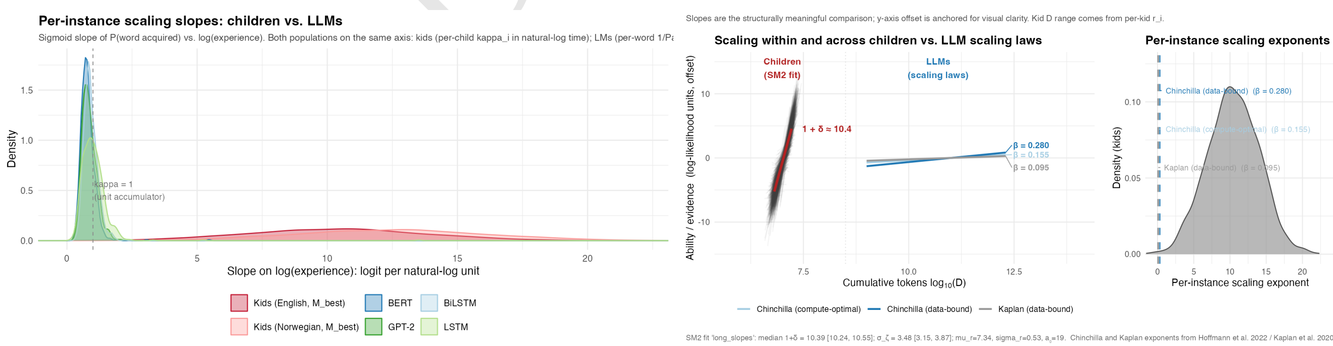


Figure 6